

10

Factorial Repeated Measures ANOVA

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Introduction

This chapter assumes that you have a good grasp of the concepts and procedures described in the chapters relevant to the factorial between-subjects ANOVA and the mixed-design ANOVA. Furthermore, if you understand the distinction between main effects, interactions, and simple main effect analyses, you have much of the knowledge essential to learn relatively easily how to conduct and interpret a factorial repeated measures ANOVA. Essentially, with one exception, and it's an important one, there are no new concepts associated with the factorial repeated measures ANOVA. The one exception is the appropriate decomposition of an interaction effect.

As with previously covered factorial ANOVAs, the most complicated element of a factorial repeated measures ANOVA is understanding and describing the nature of a significant interaction. In the factorial between-subjects and mixed-design ANOVA chapters, I wrote that a statistically significant interaction within the ANOVA summary table implies that the corresponding simple main effect Cohen's d (or Hedge's g) values are unequal. At a conceptual, and to some degree statistical, I believe this is an accessible way to understand the nature of a statistically significant interaction. However, there is a more technical and accurate way to describe and explain the nature of an interaction, which I describe in this chapter. It is especially important to describe the more accurate way to decompose an interaction effect in the factorial repeated measures ANOVA, as the use of a pooled error term that takes into consideration the shared variance across conditions becomes all the more important, in this context.

Keep in mind that a factorial repeated measures ANOVA is consistent with a fully within-subjects design. Thus, all cases provide a score across two or more independent variables, and each independent variable would be expected to be associated with two or more levels. For example, the two-way repeated measures ANOVA is considered a factorial ANOVA, because it includes two independent variables. Of course, a factorial repeated measures ANOVA is not restricted to only two independent variables. However, in this chapter, for the sake of simplicity, I cover only two-way repeated measures ANOVAs. The principles associated with conducting factorial repeated measures ANOVAs with more than two independent variables are the same.

In a study that had an impact on my life, Campbell (1990) was interested in determining whether the level of comfort students perceived in an office was influenced by whether a lecturer/professor's office had plants in it or not.¹ Furthermore, Campbell (1990) thought that whether an office was messy or tidy would also impact the degree to which

¹ After I lectured on the Campbell (1990) study the first time, a student gifted me the lily in the photo, which now lives in my office and overlooks edits to my written work.



students felt comfortable in the office. Finally, Campbell (1990) wondered whether the impact plants might have on student comfort might interact with whether the professor's office was messy or tidy. To test the hypotheses, Campbell (1990) had 251 students view four photos of the same office depicted in four different ways:² plants/tidy, plants/untidy, no plants/tidy, and no plants/untidy. Thus, there were two independent variables each with two levels: plants (yes/no) and cleanliness (tidy/untidy). Consequently, the data may be said to be consistent with a 2 x 2 repeated measures ANOVA. While viewing the photos, the students provided ratings on a scale from 1 to 9 with respect to how comfortable they would

feel, if seated as a visitor in the office. I simulated data to correspond reasonably closely to the results reported by Campbell ([Data File: plants tidy](#)). For thoroughness, I specified the key null hypotheses below:

Hypotheses

Main Effect Null Hypothesis, Plants (H_0): The office comfort mean rating means will be equal across the plant independent variable levels (yes-plants versus no-plants).

Main Effect Null Hypothesis, Cleanliness (H_0): The office comfort rating means will be equal across the cleanliness independent variable levels (tidy versus untidy).

Interaction Null Hypothesis, Plants by Cleanliness (H_0): The magnitude of the effect of the plants independent variable on office mean ratings will be equal across both levels of cleanliness.³

Next, I describe the results associated with conducting the analysis in SPSS.

Main Effects and Interaction: SPSS

In a 2x2 factorial repeated measures ANOVA, there is no possibility to violate the assumption of sphericity.⁴ Furthermore, the assumption of homogeneity of variance is of no

² Actually, Campbell (1990) was interested in additional variables, as well, however, in this example, I consider only two independent variables.

³ The interaction could also be stated as: the magnitude of the effect of the cleanliness independent variable on office comfort mean ratings will be equal across both levels of plants. These two statements are two sides of the same coin.

concern in fully within-subjects designs. The only assumptions to consider are the same as those relevant to the one-way within-subjects ANOVA. In order to conduct the factorial repeated measures ANOVA in SPSS, the General Linear Model utility can be used. When I did so, I obtained the following results (**Watch Video 10.1:** [Factorial Within-Subjects ANOVA in SPSS](#)).

As can be seen in the SPSS table entitled ‘Descriptive Statistics’, the largest numerical mean was associated with the plants/tidy condition ($M = 6.91$). By contrast, the smallest numerical mean was associated with the no plants/untidy condition ($M = 3.96$). Of course, it remains to be determined whether any of the numerical differences between the means are statistically significantly different from each other. It will be noted that the means reported in the ‘Descriptive Statistics’ table are pertinent to the hypothesized interaction effect. The means relevant to the main effect hypotheses are not reported in the ‘Descriptive Statistics’ table, unfortunately. Instead, they are reported near the end of the output, assuming the user has selected the appropriate option in SPSS.

Descriptive Statistics

	Mean	Std. Deviation	N
plants_tidy	6.908	1.495	251
plants_untidy	4.418	1.781	251
noplants_tidy	5.932	1.647	251
noplants_untidy	3.960	1.741	251

The SPSS table entitled ‘Tests of Within-Subjects Effects’ contains the key results associated with the main effect and interaction hypotheses. It can be seen that the plants main effect was significant statistically, $F(1, 250) = 140.973$, $p < .001$, partial $\eta^2 = .361$. It is not known, yet, which mean was larger than which, as SPSS reports the main effect means later in the output. The cleanliness main effect was also significant statistically, $F(1, 250) = 1596.90$, $p < .001$, partial $\eta^2 = .865$. Note that the percentage of variance in the student comfort ratings has exceeded 100% ($36.1 + 86.5 = 122.6$). The reason such an observation can occur is because the main effects do not represent unique effects: they are overlapping effects. As a general statement, main effect partial eta squared values from a factorial repeated measures ANOVA are not interpretable.

⁴ Clearly, it’s impossible to even test for sphericity for an analysis with only two levels (i.e., there is only one standard error of the difference between the means). However, I wonder whether the assumption of sphericity should apply to the results relevant to the 2x2 interaction, as the standard error of the difference between the means may not be equal across the independent variable levels. Currently, I am not familiar with any research that has been conducted to evaluate this issue. Furthermore, statistical programs do not test the assumption of sphericity, in this context. Perhaps someone will clarify this issue one day.

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It is necessary to specifically request the main effect means in a factorial repeated measures ANOVA. I obtained the results reported below in the SPSS tables entitled '1. plants' and '2. cleanliness'. It can be seen that the 'plants' in the office student comfort mean was 5.663 and the 'no-plants' in the office student comfort mean was 4.946 (i.e., lower). The difference between the two main effect means was statistically significant, as reported in the previous paragraph. Unfortunately, SPSS simply labels the means as '1' and '2'. You have to decode which mean is which, by recalling how you placed the variables into the analysis.

Next, as can be seen in the '2. cleanliness' SPSS table below, the office cleanliness main effect student comfort means were 6.420 (tidy) and 4.189 (untidy). As reported in the previous paragraph, the difference between these two means was also statistically significant. Thus, the results suggest that having an untidy office affects student comfort negatively. Whether these omnibus main effects can be interpreted straightforwardly will depend on whether the interaction was significant statistically, which I turn to next. There is also the issue of calculating a standardized measure of effect size for these mean differences (e.g., Cohen's d), an issue I return to later in the chapter.

Finally, the interaction effect was also significant statistically, $F(1, 250) = 17.31$, $p < .001$, partial $\eta^2 = .065$. In contrast to the main effects within a factorial design, the interaction effect represents a unique effect – i.e., not shared (overlapping) with the main effects. Thus, 6.5% of the variance in student comfort ratings was accounted for by the combined effect of the independent variables on the dependent variable, independently of the main effects, and independently of the shared variance across the conditions. The interaction relevant means are reported in the SPSS table entitled '3. plants * tidy', which can be found below. It will be noted that the same means are reported in the first SPSS table entitled 'Descriptive Statistics'. However, the table below also includes the 95% confidence intervals, which will prove useful for the purposes of creating a visual depiction of the interaction in a chart. Next, I demonstrate how to create the chart. Then, I describe a simple main effects-based method to determine whether the main effects can be interpreted, which is not so straightforward, given that there was a statistically significant interaction, in this example.

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Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
plants	Sphericity Assumed	129.084	1	129.084	140.973	.000	.361
	Greenhouse-Geisser	129.084	1	129.084	140.973	.000	.361
	Huynh-Feldt	129.084	1	129.084	140.973	.000	.361
	Lower-bound	129.084	1	129.084	140.973	.000	.361
Error(plants)	Sphericity Assumed	228.916	250	.916			
	Greenhouse-Geisser	228.916	250	.916			
	Huynh-Feldt	228.916	250	.916			
	Lower-bound	228.916	250	.916			
cleanliness	Sphericity Assumed	1249.402	1	1249.402	1596.9	.000	.865
	Greenhouse-Geisser	1249.402	1	1249.402	1596.9	.000	.865
	Huynh-Feldt	1249.402	1	1249.402	1596.9	.000	.865
	Lower-bound	1249.402	1	1249.402	1596.9	.000	.865
Error(cleanliness)	Sphericity Assumed	195.598	250	.782			
	Greenhouse-Geisser	195.598	250	.782			
	Huynh-Feldt	195.598	250	.782			
	Lower-bound	195.598	250	.782			
plants * cleanliness	Sphericity Assumed	16.833	1	16.833	17.306	.000	.065
	Greenhouse-Geisser	16.833	1	16.833	17.306	.000	.065
	Huynh-Feldt	16.833	1	16.833	17.306	.000	.065
	Lower-bound	16.833	1	16.833	17.306	.000	.065
Error (plants*cleanliness)	Sphericity Assumed	243.167	250	.973			
	Greenhouse-Geisser	243.167	250	.973			
	Huynh-Feldt	243.167	250	.973			
	Lower-bound	243.167	250	.973			

1. plants

Measure: MEASURE_1

plants	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	5.663	.094	5.477	5.849
2	4.946	.099	4.751	5.141

2. cleanliness

Measure: MEASURE_1

cleanliness	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	6.420	.091	6.242	6.599
2	4.189	.101	3.990	4.388

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Gignac, G. E. (2023). *How2statsbook* (Online Edition 2). Perth, Australia: Author.

3. plants * tidy

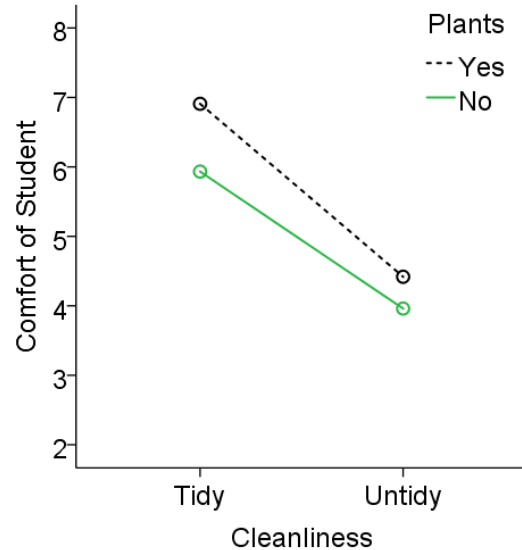
Measure: MEASURE_1

plants	tidy	Mean	Std. Error	95% Confidence Interval	
				Lower Bound	Upper Bound
1	1	6.908	.094	6.722	7.094
	2	4.418	.112	4.197	4.640
2	1	5.932	.104	5.728	6.137
	2	3.960	.110	3.744	4.177

Interaction: Graphical and Statistical Representation

Interactions are usually best represented in a graph for the purposes of description. As can be seen in Figure C10.1 (**Watch Video 10.2: [Within-Subjects ANOVA Line Chart in SPSS](#)**), the means running from tidy to untidy were not perfectly parallel. The non-parallel element of the results reflect the fact that there was a statistically significant interaction. When there is a total absence of an interaction effect (i.e., interaction $F = 1.0$), the lines connecting the corresponding means will be perfectly parallel. Of course, a non-parallel pattern in the differences between the means may arise simply due to chance, hence, the importance of testing the interaction hypothesis statistically.

Figure C10.1. Plot of Comfort Means Associated with the Interaction



From a more statistical perspective, the interaction implies that the magnitude of the difference between the 'plant' versus 'no-plant' means was unequal across the two levels of the other independent variable, i.e., 'tidy' versus 'untidy'. Correspondingly, because there are two (identical) sides to an interaction story, it may also be said that the interaction implies that

the magnitude of the difference between the ‘tidy’ versus ‘untidy’ means was unequal across the two levels of the other independent variable, i.e., ‘plants’ versus ‘no-plants’

There are several ways to represent the magnitude of the difference between two means. In the between-subjects factorial ANOVA chapter, it was recommended to represent the magnitude of the difference between means across two or more levels of another independent variable with Cohen’s d (or Hedge’s g). In my view, this is an informative method to help describe the nature of an interaction. However, in a fully within-subjects design, the use of Cohen’s d , in this context, is no longer sufficiently appropriate, because Cohen’s d fails to take into consideration the fact that the standard error of the difference between two means is reduced by a function of the shared variance across the conditions in within-subjects designs. Consequently, the magnitude of the difference between two means in the fully within-subjects factorial design needs to reflect the reduction in standard error.

In this context, it is arguably more appropriate to describe a statistically significant interaction as consistent with the observation of simple main effect t -test values that are statistically significantly different from each other. In a 2x2 factorial within-subjects ANOVA, there are two series of simple main effect paired samples t -tests that could be performed. In the current example, the two series of paired samples t -tests correspond to:

Series 1: t_1 = ‘plants tidy’ vs. ‘no plants tidy’ and t_2 = ‘plants untidy’ vs. ‘no plants untidy’

Series 2: t_3 = ‘plants tidy’ vs. ‘plants untidy’ and t_4 = ‘no plants tidy’ vs. ‘no plants untidy’

To repeat, a statistically significant interaction implies that the two paired sample t -tests within a series were statistically significantly different from each other (**Watch Video 10.3: [Within-Subjects Interactions, Cohen’s \$d\$ vs. \$t\$ -values](#)**). It is tempting to simply run regular paired samples t -tests, in this context. However, if one did so, each of the paired samples t -tests would be based on a separately estimated standard error of the difference between the means error term. Separately estimated standard errors is inconsistent with the 2x2 interaction tested by the ANOVA, because the ANOVA was based on a *pooled* standard error term. Correspondingly, in the ANOVA table, the interaction error term is based on the mean square error (MS_E) associated with the interaction. In this example, the interaction MS_E was estimated at .973, as reported in the above SPSS table entitled ‘Tests of Within-Subjects Effects’.

In order to convert the MS_E from the ANOVA into a pooled standard error of the difference between related means term, the following formula can be used (**Watch Video 10.4: [Within-Subjects ANOVA Pooled Error Term](#)**):

$$\text{pooled } SE_D = \sqrt{\frac{MS_E}{(N/2)}} \quad (1)$$

In the current example, the pooled SE_D was estimated at:

$$\text{pooled } SE_D = \sqrt{\frac{.973}{(251/2)}} = .088$$

Thus, in this case, the ‘proper’ simple main effect paired samples t -tests associated with the factorial ANOVA can be performed with the estimate of .088 as the standard error of the difference between two related means:

$$\text{pooled } SE_D t_1 = \frac{6.908 - 5.932}{.088} = 11.091 \quad \text{pooled } SE_D t_2 = \frac{4.418 - 3.960}{.088} = 5.205$$

The difference between the two pooled error paired samples t -tests amounted to 5.886 (i.e., 11.091 – 5.205). Thus, the statistically significant 2x2 interaction reported in the ANOVA summary table implies that the 11.091 and 5.205 t -values were statistically significantly different from each other. Stated alternatively, the difference in t -values of 5.886 was statistically significantly different from zero. When the interaction null hypothesis is true, the difference between the two t -values will be zero, within sampling fluctuations. Thus, the magnitude of the effect of having plants in a tidy office, in comparison to no plants in a tidy office, was statistically significant greater than the magnitude of the effect of having plants in an untidy office, in comparison to no plants in an untidy office. Such an interpretation is consistent with the results presented in Figure 10.1. In lay person terms, the positive effect of having plants in an office is statistically significantly greater, when the office is tidy relative to untidy. Stated alternatively (again), the magnitude of the effect of the presence of plants (vs no plants) on student comfort in an office varies as a function of the cleanliness of the office.

The above two pooled paired samples t -tests represent one perspective of the interaction effect. As alluded to earlier, a 2x2 interaction can be interpreted from two perspectives, in the same way that a coin has two sides. In this example, it would be perfectly legitimate to interpret the interaction from the perspective of the difference between the tidy and untidy means for the plant condition versus the difference between tidy and untidy means for the no plants condition. Thus, the following pooled paired sample t -test t -values can also be compared:

$$\text{pooled } SE_D t_1 = \frac{6.908 - 4.418}{.088} = 28.295 \quad \text{pooled } SE_D t_2 = \frac{5.932 - 3.960}{.088} = 22.409$$

Again, the statistically significant interaction reported in the ANOVA summary table implies that the 28.295 and 22.409 simple main effect paired samples t -test values were statistically significantly different from each other. Importantly, the numerical difference between the pooled paired t -test values corresponded to 5.886 (28.295 – 22.409), which is identical to the difference associated with the (pooled) paired samples t -test values obtained

from the preceding analysis/perspective of the interaction. That's not a coincidence. It's the same story from a different angle.

It should be said that it is not *necessary* to conduct these follow-up pooled paired samples *t*-tests, in order to test an interaction hypothesis. Many researchers simply describe the nature of an interaction with reference to a plot of means in a chart. The material is presented here to help students and researchers alike understand the true nature of an interaction, as there appears to be a fair amount of confusion on this issue, particularly given the manner in which one sees simple main effect analyses reported in the literature. As I have described in the factorial between-subjects ANOVA chapter, the statistical significance associated with the individual simple main effect analysis *t*-values are not relevant to the testing and/or interpreting interaction hypotheses. Instead, it's the difference between the *t*-values that is of importance. Correspondingly, the statistical significance associated with each of the simple main effect analyses are pertinent to the potential interpretation of an omnibus main effect in the presence of a statistically significant interaction. I turn to this issue next.

Simple Main Effect Analyses

In the factorial between-subjects ANOVA chapter, it was mentioned that when there is a statistically significant omnibus⁵ main effect, and there is a statistically significant omnibus interaction, the main effect may be, or may not be, interpretable. In order to determine whether a statistically significant main effect can be interpreted, in the presence of a statistically significant interaction, a series of simple main effect analyses need to be performed. Specifically, the simple main effect analyses need to be statistically significant across all levels of the interacting independent variable, and in the same direction, in order to interpret the main effect justifiably (Franz & Loftus, 2012; Loftus & Masson, 1994).

There are two approaches to conducting simple main effect analyses in the factorial repeated measures ANOVA case: (1) the quick and dirty way with conventional paired samples *t*-tests; and (2) paired sample *t*-test contrasts with a pooled error term. In practice, most researchers apply the quick and dirty method. In most cases, I suspect most reviewers of your work will let you get away with the quick and dirty method. However, it is worthwhile taking the time to understand the more accurate approach, based on the pooled error term, for two reasons. First, in practice, the results will be more accurate. Secondly, the pooled error term

⁵ Recall that an omnibus effect, in this context, represents a statistical result that amalgamates two or more hypotheses, simultaneously. In a 3x3 factorial design, the main effect and interaction effect ANOVA *F*-values, for example, potentially represent multiple sources of systematic variance that may need to be decomposed with follow-up analyses. Technically, the main effects within a 2x2 factorial design are not truly omnibus in nature, as each main effect necessarily represents the difference between only two means. However, the possibility of a statistically significant interaction between the two independent variables does add a dimension to the main effects. Consequently, the main effects from a 2x2 design may be considered omnibus from that perspective.

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forms the foundation for understanding the true nature of an interaction from a statistical perspective, which I described in some detail in the preceding section of this chapter. In the next section of this chapter, I describe the quick and dirty method to conducting simple main effect analyses. I also describe a method to create a factorial repeated measures ANOVA chart with conventional 95% confidence intervals in SPSS. In the Advanced Topics section of this chapter, I describe in detail the more accurate pooled error simple main effects method with statistical significance testing, as well as a more accurate method to create 95% confidence intervals for a factorial repeated measures ANOVA plot of means.

Simple Main Effect Analyses: Separate Error Terms in SPSS

In the factorial between-subjects ANOVA chapter, simple main effect analyses consisted of a series of isolated independent sample *t*-tests conducted across all levels of the interacting independent variable. In the factorial repeated measures ANOVA case, the same procedure applies, except that a series of paired samples *t*-tests need to be conducted at each level of the interacting independent variable.

First, I will follow-up the statistically significant omnibus ‘plant’ main effect, which, you may recall, was significant statistically, $F(1, 250) = 140.97$, $p < .001$. With respect to the plant main effect, the interacting variable was cleanliness, which had two levels: tidy and untidy. Thus, two simple main effect paired *t*-tests need to be performed: (1) ‘plants’ versus ‘no plants’, isolated at ‘tidy’ office; and (2) ‘plants’ versus ‘no plants’, isolated at ‘untidy’ office. If both simple main effect paired *t*-tests are found to be significant statistically, and they are in the same direction, the main effect of plants can be interpreted justifiably.

In order to conduct the analysis in SPSS, the ‘Analyze = Compare Means’ menu utility can be used (**Watch Video 10.5: [Within-Subjects Factorial ANOVA Simple Main Effects in SPSS](#)**). As can be seen in the SPSS table entitled ‘Paired Samples Test’, both simple main effect paired samples *t*-tests were significant statistically ($p < .001$). Furthermore, the mean difference was in the same direction for both analyses (i.e., .976 and .458). Thus, the ‘plants’ omnibus main effect can be interpreted. In plain language, the inclusion of plants in a professor’s office was associated with higher mean student comfort ratings, irrespective of whether the office was tidy or untidy.

Paired Samples Test									
		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	plants_tidy - noplants_tidy	.976	1.284	.081	.817	1.136	12.048	250	.000
Pair 2	plants_untidy - noplants_untidy	.458	1.459	.092	.277	.640	4.974	250	.000

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Next, I follow-up the statistically significant omnibus ‘cleanliness’ main effect, which, you may recall, was also significant statistically, $F(1, 250) = 1596.90$, $p < .001$. With respect to the cleanliness main effect, the interacting variable was office plants, which had two levels: ‘plants’ and ‘no plants’. Thus, two simple main effect paired t -tests need to be performed: (1) ‘tidy’ versus ‘untidy’, isolated at ‘plants’ in the office; and (2) ‘tidy’ versus ‘untidy’, isolated at ‘no plants’ in the office. If both simple main effect paired t -tests are found to be significant statistically, and they are in the same direction, the main effect of cleanliness can be interpreted justifiably.

As can be seen in the SPSS table entitled ‘Paired Samples Test’, both simple main effect paired samples t -tests were significant statistically ($p < .001$). Furthermore, the mean difference was in the same direction for both analyses (i.e., 2.490 and 1.972). Thus, the ‘cleanliness’ omnibus main effect can be interpreted. In plain language, a tidy office was associated with higher mean comfort ratings by students, irrespective of whether there were plants in the office or not.

Paired Samples Test									
		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	plants_tidy - plants_untidy	2.490	1.366	.086	2.320	2.660	28.872	250	.000
Pair 2	noplants_tidy - noplants_untidy	1.972	1.282	.081	1.813	2.131	24.374	250	.000

Allow me to repeat: these follow-up simple main effect analyses are only necessary, if a statistically significant omnibus main effect is observed in the presence of a statistically significant omnibus interaction. If there is no omnibus statistically significant interaction, a statistically significant omnibus main effect can be interpreted on its own. It is not necessary to follow-up with the simple main effect analyses.

One thing you may not have noticed across the four simple main effect paired sample t -tests that were performed in this section is that they were each associated with a different standard error of the difference between two related means (see ‘Std. Error Mean’ column in the SPSS tables). Specifically, the first two paired t -tests were associated with standard errors of .081 and .092. The second lot of paired t -tests were associated with standard errors of .086 and .081. The fact that each paired t -test was associated with its own standard error is inconsistent with the fact that an omnibus main effect is based on a pooled error term, which is labeled ‘Mean Square’ (column) ‘Error’ (row) in the ANOVA summary table. For example, with respect to the ‘plants’ main effect, the mean square error was reported at .916. Statistically, it is more appropriate (i.e., accurate) to conduct the simple main effect paired t -tests, based on a pooled standard error of the difference between two related means term. As mentioned, I believe most people (reviewers, supervisors, carpenters, etc.) will let you get away with conducting the simple main effect paired samples t -tests based on separate error

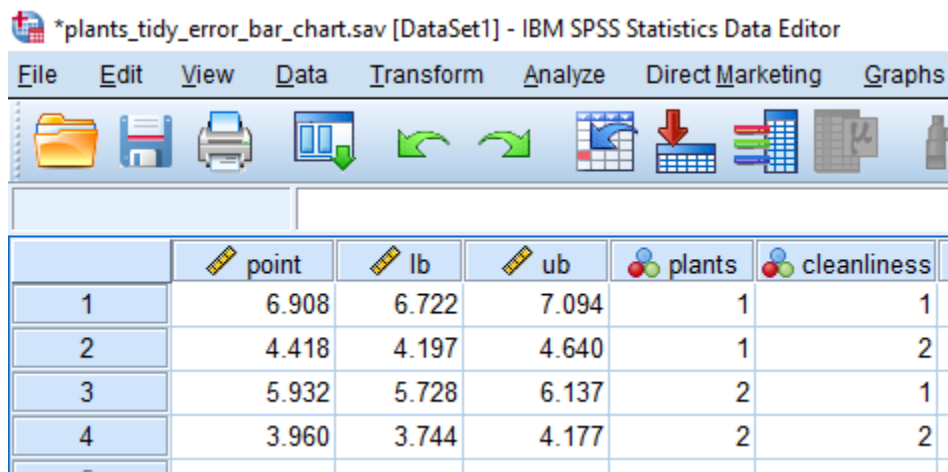
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terms. However, for those of you keen on conducting simple main effect analyses properly with a pooled error term, I encourage you to consult the relevant portion of the Advanced Topics section of this chapter.

Plotting Factorial Repeated Measures ANOVA Means with 95% CIs

Ideally, there would be a way to create the plot of means presented in Figure C10.1 which includes the 95% confidence intervals in SPSS. If you have access to SPSS Version 25 or above, there is an easy way to do it. You simply have to use the ‘Plots’ option within the factorial within-subjects ANOVA analysis itself (**Watch Video 10.6: [Within-Subjects Factorial ANOVA Chart with CIs in SPSS - Easy](#)**). However, if you have access to earlier versions of SPSS, there is a less easy, but still manageable, way to do it. The work-around method is to create a separate SPSS data file and input the means (point), lower-bounds (lb), and upper-bound (ub) confidence intervals. You also need to identify the means with the two levels on the first within-subjects factor (plants) and the two levels on the second within-subjects factor (cleanliness). Below, I have depicted the data file I created for the current example.



	point	lb	ub	plants	cleanliness
1	6.908	6.722	7.094	1	1
2	4.418	4.197	4.640	1	2
3	5.932	5.728	6.137	2	1
4	3.960	3.744	4.177	2	2

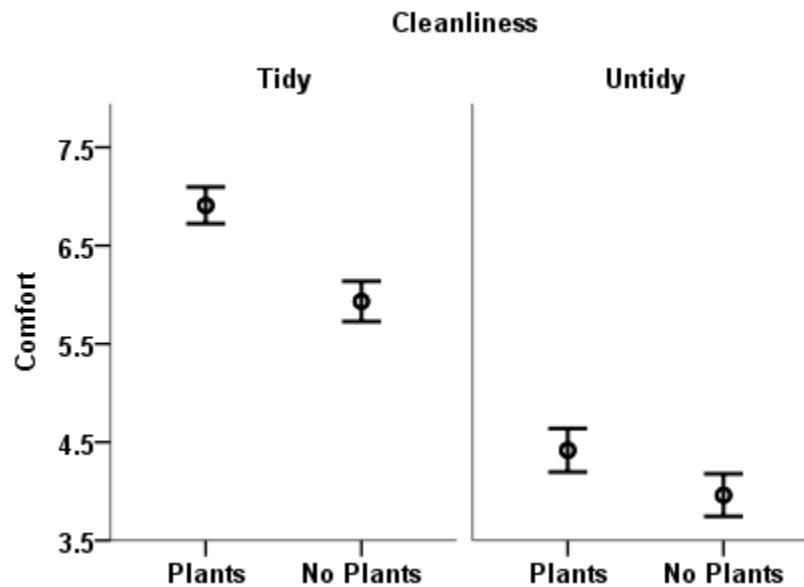
Once the data are inputted into the data editor, you can run a ‘High-Low’ chart in SPSS (**Watch Video 10.7: [Within-Subjects Factorial ANOVA Chart with CIs in SPSS – Less Easy](#)**). The chart I created can be found in Figure C10.2. It can be observed that the distance between the ‘plants’ versus ‘no plants’ means was larger for the ‘Tidy’ condition, in comparison to the distance between ‘plants’ versus ‘no plants’ means for the ‘untidy’ condition. Such a pattern in the means is consistent with the statistically significant interaction observed in the ANOVA summary table. Basically, adding a plant to a professor’s office has a statistically significantly larger impact on student comfort, when the office is tidy, in comparison to untidy.

It should be noted that the above 95% confidence intervals were obtained from what SPSS outputs in the typical factorial repeated measures ANOVA. Arguably, these 95% confidence intervals are not accurate within the context of a within-subjects design ANOVA. In

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fact, these 95% confidence intervals are simply based on multiplying the standard error of each mean by 1.96. However, it has been argued (e.g., Franz & Loftus, 2012; Loftus & Masson, 1994) that the standard errors should be adjusted to reflect the fact that a factorial repeated measures ANOVA controls for the shared variance across the conditions (i.e., the usually positive correlations between the scores across the levels). If you are interested in obtaining the adjusted 95% confidence intervals that are consistent with the actual interaction hypothesis tested by the interaction ANOVA *F*-test, check out the relevant Advanced Topics section of this chapter.

Figure C10.2. Plot of Comfort Means and 95% CIs Associated with the Interaction



2 x *k* Factorial Repeated Measures ANOVA

The process of analyzing data with more than two levels associated with one or more of the within-subjects factor levels is essentially the same as that of a 2 x 2 design. The main difference resides in decomposing the interaction, if a statistically significant interaction is observed. Consistent with other 2 x *k* factorial ANOVAs, a significant interaction can often be best decomposed by performing a series of contiguous 2 x 2 factorial repeated measures ANOVAs.

For example, Gignac, Darbyshire, and Ooi (2018) had 382 young adults rate (on a 6-point scale) seven intelligence percentiles (1st, 10th, 25th, 50th, 75th, 90th, and 99th) across two conditions: sexually attractiveness specifically and (romantic) partner desirability more generally. Gignac et al. (2018) hypothesized that both the sexual attractiveness and partner desirability ratings would increase across the intelligence percentiles. That is, higher intelligence in a person is both more sexually attractive and desirable in a partner more

generally, right? Although not hypothesized specifically, there was the possibility that the sexually attractiveness rated mean (collapsed across all IQ percentiles) would be different to the partner desirability rated mean (collapsed across all IQ percentiles). Thus, as there were two independent variables, IQ percentile (independent variable 1) and relationship type (independent variable 2), there were two main effects. Furthermore, the degree of change across the seven intelligence percentiles might be different across the two types of relationships: sexual attractiveness specifically versus partner desirability. Thus, there was also hypothesized the possibility of an interaction. I have specified the three hypotheses more formally below:

Hypotheses

Main Effect Null Hypothesis, Relationship Type (H_0): The mean ratings (collapsed across all seven IQ percentiles) will be equal across the two relationship types (sexual attraction specifically vs. romantic partnership).

Main Effect Null Hypothesis, IQ Percentile Level (H_0): The mean ratings (collapsed across the two relationship types) will be equal across the seven IQ percentiles.

Interaction Effect Null Hypothesis, Relationship Type by IQ Percentile Level (H_0): The magnitude of the effect of IQ percentiles on mean ratings will be equal across the two relationship types (sexual attraction specifically versus romantic partnership).

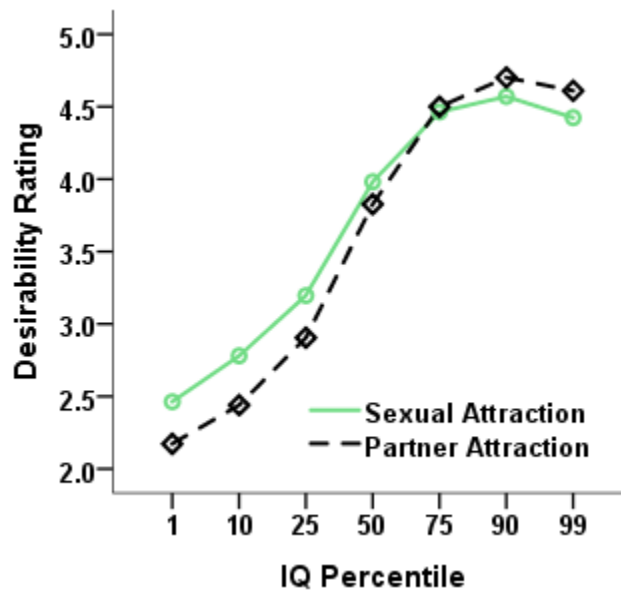
As mentioned above, the process of conducting a $2 \times k$ factorial repeated measures ANOVA is the same as that demonstrated above for the 2×2 factorial repeated measures ANOVA. Thus, based on the relevant data ([Data File: IQ desirability](#)), I conducted a 2×7 factorial repeated measures ANOVA. I think the best way to digest the results of a k by k repeated measures factorial ANOVA is to first look at the plot of means. As can be seen in Figure C10.3, both the sexual attraction specifically and partner attraction means tended to increase from the 1st percentile to the 90th percentile. However, there was at least a numerical decrease in the means from the 90th to the 99th percentiles. Thus, at least superficially, it would appear that the 90th percentile was rated the most attractive from both the perspective of sexual attraction specifically and partner desirability more generally. Of course, any numerical differences would have to be tested statistically, prior to interpretation. It is also worth stating that the pattern of means appears to be non-linear in nature. Specifically, it looks like there might be cubic trend in the means (two bends), which could be tested for statistical significance. Based on an examination of Figure 10.2, it is more difficult to tell whether there is a statistically significant main effect associated with the relationship type

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within-subjects factor (sexual attraction versus partner attraction), or whether there is an interaction between relationship type and IQ percentile level on ratings. Of course, these numerical differences could be tested for statistical significance, as well, which I did next.

In order to conduct a $k \times k$ factorial repeated measures ANOVA, one can use the General Linear Model menu option in SPSS (**Watch Video 10.8: [Factorial Within-Subjects ANOVA - Trend Analyses](#)**), as per the 2×2 factorial repeated measures ANOVA I performed in the preceding section. When I did so with the Gignac et al. (2018) data, I obtained a substantial amount of output, not all of which is relevant to the hypotheses of interest. First, as can be seen in the SPSS table entitled 'Descriptive Statistics', the sexual attractiveness and partner desirability means tended to increase in a linear fashion, at least to some degree. For example, the 1st IQ percentile for the sexual attractiveness condition (smarter_sex_1p) was associated with a mean of 2.46. By comparison, the 99th IQ percentile for the sexual attractiveness condition (smarter_sex_99p) was associated with a mean of 4.42. A similar numerical difference was observed for the partner desirability means: 1st IQ percentile (smarter_partner_1p) = 2.17 and 99th IQ percentile (smarter_partner_99p) = 4.61.

Figure C10.3. Plot of Means Associated with the Rated IQ Percentiles



Descriptive Statistics

	Mean	Std. Deviation	N
smarter_sex_1p	2.46	1.289	382
smarter_sex_10p	2.77	1.234	382
smarter_sex_25p	3.20	1.077	382
smarter_sex_50p	3.98	.834	382
smarter_sex_75p	4.46	.812	382
smarter_sex_90p	4.57	1.057	382
smarter_sex_99p	4.42	1.277	382
smarter_partner_1p	2.17	1.305	382
smarter_partner_10p	2.44	1.263	382
smarter_partner_25p	2.91	1.293	382
smarter_partner_50p	3.83	1.005	382
smarter_partner_75p	4.50	.895	382
smarter_partner_90p	4.70	1.065	382
smarter partner 99p	4.61	1.279	382

Next, as can be seen in the table entitled 'Mauchly's Test of Sphericity', the assumption of sphericity was not satisfied for the IQ percentile (IQ_level) within-subjects factor, $\chi^2(20) = 2446.29$, $p < .001$ or the relationship type by IQ percentile interaction effect, $\chi^2(20) = 578.43$, $p < .001$. No dramas. I'll simply consult the Huynh-Feldt adjusted results in the ANOVA table: Such adjusted results do not assume sphericity.

Mauchly's Test of Sphericity^a

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon ^b		
					Greenhouse-Geisser	Huynh-Feldt	Lower-bound
type	1.000	.000	0	.	1.000	1.000	1.000
IQ_level	.002	2446.287	20	.000	.284	.285	.167
type * IQ_level	.217	578.429	20	.000	.616	.623	.167

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept

Within Subjects Design: type + IQ_level + type * IQ_level

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Next, the key results associated with the analysis are reported. As can be seen in the SPSS table entitled 'Tests of Within-Subjects Effects', the relationship type main effect null hypothesis was rejected, $F(1, 381) = 23.42$, $p < .001$, partial $\eta^2 = .058$. The means associated with this rejected null hypothesis are not reported by SPSS, unless you made the necessary selections. If you did make the necessary selections, the means will be reported in a table entitled '1. type Estimates'. In this example, the means were 3.696 and 3.594 for the sexual attraction specifically and romantic partnership conditions, respectively. Thus, on average, the

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sexual attraction rated IQ percentiles were associated with slightly higher ratings than the partner desirability IQ percentile ratings. The partial eta squared value of .058 implies that 5.8% of the variance in ratings was accounted for by relationship type, controlling for the shared variance across the levels (keep in mind that there were substantial positive correlations across the ratings).

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
type	Sphericity Assumed	14.038	1	14.038	23.418	.000	.058
	Greenhouse-Geisser	14.038	1.000	14.038	23.418	.000	.058
	Huynh-Feldt	14.038	1.000	14.038	23.418	.000	.058
	Lower-bound	14.038	1.000	14.038	23.418	.000	.058
Error(type)	Sphericity Assumed	228.390	381	.599			
	Greenhouse-Geisser	228.390	381.000	.599			
	Huynh-Feldt	228.390	381.000	.599			
	Lower-bound	228.390	381.000	.599			
IQ_level	Sphericity Assumed	4355.093	6	725.849	410.856	.000	.519
	Greenhouse-Geisser	4355.093	1.704	2555.212	410.856	.000	.519
	Huynh-Feldt	4355.093	1.711	2544.940	410.856	.000	.519
	Lower-bound	4355.093	1.000	4355.093	410.856	.000	.519
Error(IQ_level)	Sphericity Assumed	4038.622	2286	1.767			
	Greenhouse-Geisser	4038.622	649.375	6.219			
	Huynh-Feldt	4038.622	651.996	6.194			
	Lower-bound	4038.622	381.000	10.600			
type * IQ_level	Sphericity Assumed	54.344	6	9.057	41.558	.000	.098
	Greenhouse-Geisser	54.344	3.695	14.707	41.558	.000	.098
	Huynh-Feldt	54.344	3.736	14.547	41.558	.000	.098
	Lower-bound	54.344	1.000	54.344	41.558	.000	.098
Error(type*IQ_level)	Sphericity Assumed	498.227	2286	.218			
	Greenhouse-Geisser	498.227	1407.80	.354			
	Huynh-Feldt	498.227	1423.30	.350			
	Lower-bound	498.227	381.000	1.308			

1. type

Estimates

Measure: MEASURE_1

type	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	3.696	.033	3.632	3.760
2	3.594	.035	3.525	3.662

Next, as can be seen in the SPSS table entitled 'Tests of Within-Subjects Effects', the main effect of intelligence percentile was also significant statistically, $F(1.711, 651.996) = 410.856$, $p < .001$, partial $\eta^2 = .519$. Thus, 51.9% of the variance in the ratings was accounted for by intelligence percentile level, controlling for the shared variance between the percentile intelligence ratings. The means associated with this main effect null hypothesis are reported

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by SPSS in a table entitled '2. IQ_level Estimates'. It can be seen that, for an intelligence level at the 1st percentile, the mean level of desirability was 2.318 (collapsed across the sexual attraction specifically and partner desirability conditions). By contrast, at the 99th intelligence percentile, the mean level of desirability was 4.517 (again, collapsed across the sexual attraction specifically and partner desirability conditions). Because there are so many means associated with this main effect analysis, it is not efficient to conduct an exhaustive series of multiple comparisons. Instead, given the pattern visible in Figure C10.3, it would be best to report whether there was curvilinear effect. In particular, it appears that there may be a cubic effect.

2. IQ_level

Estimates

Measure: MEASURE_1

IQ_level	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	2.318	.063	2.194	2.442
2	2.607	.060	2.488	2.726
3	3.051	.057	2.938	3.164
4	3.904	.043	3.820	3.989
5	4.482	.040	4.402	4.561
6	4.636	.052	4.535	4.738
7	4.517	.063	4.393	4.641

In fact, as can be seen in the SPSS Table entitled 'Tests of Within-Subjects Contrasts', the intelligence level (IQ_level) main effect was associated with a statistically significant cubic effect, $F(1, 381) = 246.563$, partial $\eta^2 = .393$. Admittedly, all of the curvilinear effects were significant statistically. However, next to the linear effect, the cubic effect was associated with the largest effect size (partial $\eta^2 = .393$). Thus, I would suggest that the best way to interpret the intelligence percentile main effect is to suggest that there is an increasing trend in the rated means, but with two important bends in the pattern. First, there was disproportionately large increase from the 10th to the 50th IQ percentiles in rated desirability, which caused one bend. The other bend in the means was consistent with an inverted-U shape from the 75th to the 99th IQ percentiles (see Figure 10.3). As I describe below, there was a statistically significant interaction, which does complicate matters.

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Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	type	IQ_level	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
type	Linear		14.038	1	14.038	23.418	.000	.058
Error(type)	Linear		228.390	381	.599			
IQ_level		Linear	3985.055	1	3985.055	527.497	.000	.581
		Quadratic	148.491	1	148.491	105.557	.000	.217
		Cubic	202.306	1	202.306	246.563	.000	.393
		Order 4	2.869	1	2.869	8.399	.004	.022
		Order 5	13.915	1	13.915	54.915	.000	.126
		Order 6	2.457	1	2.457	11.010	.001	.028
Error(IQ_level)		Linear	2878.320	381	7.555			
		Quadratic	535.967	381	1.407			
		Cubic	312.611	381	.821			
		Order 4	130.154	381	.342			
		Order 5	96.543	381	.253			
		Order 6	85.026	381	.223			
type * IQ_level	Linear	Linear	49.305	1	49.305	91.508	.000	.194
		Quadratic	1.666	1	1.666	7.657	.006	.020
		Cubic	3.194	1	3.194	18.516	.000	.046
		Order 4	.005	1	.005	.043	.837	.000
		Order 5	.141	1	.141	1.215	.271	.003
		Order 6	.033	1	.033	.240	.625	.001
Error(type*IQ_level)	Linear	Linear	205.284	381	.539			
		Quadratic	82.911	381	.218			
		Cubic	65.723	381	.173			
		Order 4	47.459	381	.125			
		Order 5	44.103	381	.116			
		Order 6	52.747	381	.138			

Finally, the interaction null hypothesis was also rejected, $F(3.736, 1423.30) = 41.558$, $p < .001$, partial $\eta^2 = .098$. Thus, the manner in which the means increased from the 1st percentile to the 99th percentile differed across the two relationship types: sexual attraction specifically versus partner desirability.

In order to help uncover the nature of the interaction, statistically, one may conduct a series of 2 x 2 factorial repeated measures ANOVAs. As there were seven levels associated with the intelligence percentile within-subjects factor, there are six possible contiguous (i.e., side by side) 2 x 2 factorial repeated measures ANOVAs. As can be seen in Table C10.1, the only statistically significant 2 x 2 interaction effects were observed across the 25th, 50th, 75th, and 90th percentiles. Additionally, the 2 x 2 factorial repeated measures ANOVA that compared the 25th and 90th percentiles (i.e., non-contiguous) was associated with an effect ($\eta^2 = .133$, $p < .001$) size noticeably larger than the omnibus effect size (i.e., $\eta^2 = .098$). In lay person terms, the statistically significant interaction implied that, across the 25th to 90th percentiles of intelligence, the impact of intelligence level on desirability rating for the prospective partner condition was much more substantial than the sexual attraction specifically condition.

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Table C10.1. Contiguous 2 x 2 Factorial Repeated Measures ANOVAs

Comparison	Partial η^2	p-value
1 st vs. 10 th	.005	.167
10 th vs. 25 th	.003	.249
25 th vs. 50 th	.023	.003
50 th vs. 75 th	.052	< .001
75 th vs. 90 th	.018	.009
90 th vs. 99 th	.009	.064

Simple Main Effects

Given the statistically significant interaction, more analyses are required to potentially interpret the omnibus main effects, both of which were significant statistically (relationship type and intelligence percentile). As per the other factorial ANOVA chapters, what is required, here, are two series of simple main effect analyses. Recall that, in a factorial design with two independent variables, simple main effect analyses consist of mean comparisons isolated at each level of the other independent variable. For example, in order to interpret the relationship type main effect, the sexual attraction specifically versus romantic partnership mean comparisons (simple main effects) must be significant statistically across all seven intelligence percentiles. Furthermore, all of the significant effects must be in the same direction, as well. Additionally, in order to interpret the intelligence percentile main effect, the mean intelligence percentile rating comparisons across contiguous percentiles (e.g., 10th versus 25th) must be significant statistically for both the sexually attractive specifically and romantic partner conditions.

As per previous factorial ANOVA chapters, the quick and dirty approach would involve conducting a series of paired samples *t*-tests on the two relationship type means across all seven intelligence percentiles. Many reviewers would probably let you get away with this approach, even though it is not entirely accurate, as each paired *t*-test would be based on a separately estimated standard error of the difference between means. Recall that the omnibus main effect is based on a pooled standard error (mean square error). A more sophisticated approach would involve conducting the mean comparisons, based on common (pooled) standard error. Unfortunately, to my knowledge, it is not possible to conduct such simple main effect analyses in SPSS, when there is a within-subjects factor. If you are ambitious, I demonstrate a method (“by hand”) to conduct simple main effect analyses with a pooled error term in the Advanced Topics section of this chapter.

If you opt for the quick and dirty approach, as many people do, it is useful to get SPSS to conduct the simple main effect analyses efficiently by adjusting the relevant syntax slightly to produce the non-pooled simple main effect analyses in a single table, with or without a Bonferroni correction (**Watch Video 10.9:** [More Within-Subjects Factorial ANOVA Simple Main](#)

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[Effects in SPSS](#)). The adjustment involves adding the COMPARE command to the syntax, as can be seen below.

```
GLM smarter_sex_1p smarter_sex_10p smarter_sex_25p smarter_sex_50p smarter_sex_75p
smarter_sex_90 smarter_sex_99p smarter_partner_1p smarter_partner_10p
smarter_partner_25p smarter_partner_50p smarter_partner_75p smarter_partner_90p
smarter_partner_99p
/WSFACTOR=type 2 Polynomial IQ_level 7 Polynomial
/METHOD=SSTYPE(3)
/EMMEANS=TABLES(type*IQ_level) COMPARE (type)
/PRINT=DESCRIPTIVE
/CRITERIA=ALPHA(.05)
/WSDESIGN=type IQ_level type*IQ_level.
```

When I ran the syntax above, I obtained the key SPSS Table entitled ‘Pairwise Comparisons’. It can be seen that not all of the mean comparisons were significant statistically. Specifically, the comparison at IQ_level 5, which corresponds to IQ percentile 75, was not significant, $p = .272$. Furthermore, some of the comparisons were positively significant (i.e., larger sexually attractive mean than partner desirability mean) and others were negatively significant (i.e., larger partner desirability mean than sexually attractive specifically mean). Thus, the significant omnibus main effect of relationship type cannot be interpreted.

Pairwise Comparisons

Measure: MEASURE_1

IQ_level	(I) type	(J) type	Mean Difference (I-J)	Std. Error	Sig. ^b	95% Confidence Interval for Difference ^b	
						Lower Bound	Upper Bound
1	1	2	.291 [*]	.041	.000	.210	.371
	2	1	-.291 [*]	.041	.000	-.371	-.210
2	1	2	.335 [*]	.041	.000	.254	.416
	2	1	-.335 [*]	.041	.000	-.416	-.254
3	1	2	.291 [*]	.041	.000	.211	.371
	2	1	-.291 [*]	.041	.000	-.371	-.211
4	1	2	.154 [*]	.039	.000	.077	.232
	2	1	-.154 [*]	.039	.000	-.232	-.077
5	1	2	-.037	.033	.272	-.102	.029
	2	1	.037	.033	.272	-.029	.102
6	1	2	-.131 [*]	.034	.000	-.198	-.064
	2	1	.131 [*]	.034	.000	.064	.198
7	1	2	-.186 [*]	.034	.000	-.252	-.120
	2	1	.186 [*]	.034	.000	.120	.252

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

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Gignac, G. E. (2023). *How2statsbook* (Online Edition 2). Perth, Australia: Author.

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With respect to the IQ percentile level main effect, it is clear that the main effect cannot be interpreted, as the means are associated with a curvilinear effect, as described earlier in the chapter. However, with a different set of data, one would consider testing the simple main effect analyses, in order to determine whether the second main effect could be interpreted. Again, you would be required to modify the corresponding SPSS syntax by including the relevant COMPARE command, as can be seen below. For thoroughness, I have reported the corresponding simple main effect results in the SPSS table entitled 'Pairwise Comparisons'.

```
GLM smarter_sex_1p smarter_sex_10p smarter_sex_25p smarter_sex_50p smarter_sex_75p  
smarter_sex_90p  
    smarter_sex_99p    smarter_partner_1p    smarter_partner_10p    smarter_partner_25p  
smarter_partner_50p  
    smarter_partner_75p smarter_partner_90p smarter_partner_99p  
/WSFACTOR=type 2 Polynomial IQ_level 7 Polynomial  
/METHOD=SSTYPE(3)  
/EMMEANS=TABLES(type*IQ_level) COMPARE (IQ_level)  
/CRITERIA=ALPHA(.05)  
/WSDESIGN=type IQ_level type*IQ_level.
```

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Pairwise Comparisons

Measure: MEASURE_1

type	(I) IQ_level	(J) IQ_level	Mean Difference (I-J)	Std. Error	Sig. ^b	95% Confidence Interval for Difference ^b	
						Lower Bound	Upper Bound
1	1	2	-.312 [*]	.033	.000	-.376	-.247
		3	-.733 [*]	.046	.000	-.823	-.643
		4	-1.518 [*]	.067	.000	-1.650	-1.387
		5	-2.000 [*]	.082	.000	-2.162	-1.838
		6	-2.107 [*]	.095	.000	-2.295	-1.920
		7	-1.961 [*]	.101	.000	-2.159	-1.763
	2	1	.312 [*]	.033	.000	.247	.376
		3	-.421 [*]	.032	.000	-.484	-.359
		4	-1.207 [*]	.060	.000	-1.325	-1.089
		5	-1.688 [*]	.079	.000	-1.844	-1.533
		6	-1.796 [*]	.093	.000	-1.979	-1.613
		7	-1.649 [*]	.099	.000	-1.843	-1.455
	3	1	.733 [*]	.046	.000	.643	.823
		2	.421 [*]	.032	.000	.359	.484
		4	-.785 [*]	.047	.000	-.877	-.694
		5	-1.267 [*]	.067	.000	-1.398	-1.136
		6	-1.374 [*]	.082	.000	-1.536	-1.212
		7	-1.228 [*]	.091	.000	-1.406	-1.049
	4	1	1.518 [*]	.067	.000	1.387	1.650
		2	1.207 [*]	.060	.000	1.089	1.325
		3	.785 [*]	.047	.000	.694	.877
		5	-.482 [*]	.042	.000	-.564	-.399
		6	-.589 [*]	.064	.000	-.714	-.464
		7	-.442 [*]	.076	.000	-.592	-.293
	5	1	2.000 [*]	.082	.000	1.838	2.162
		2	1.688 [*]	.079	.000	1.533	1.844
		3	1.267 [*]	.067	.000	1.136	1.398
		4	.482 [*]	.042	.000	.399	.564
		6	-.107 [*]	.041	.009	-.188	-.027
		7	.039	.059	.504	-.076	.155
	6	1	2.107 [*]	.095	.000	1.920	2.295
		2	1.796 [*]	.093	.000	1.613	1.979
		3	1.374 [*]	.082	.000	1.212	1.536
		4	.589 [*]	.064	.000	.464	.714
		5	.107 [*]	.041	.009	.027	.188
		7	.147 [*]	.037	.000	.074	.219
	7	1	1.961 [*]	.101	.000	1.763	2.159
		2	1.649 [*]	.099	.000	1.455	1.843
		3	1.228 [*]	.091	.000	1.049	1.406
		4	.442 [*]	.076	.000	.293	.592
		5	-.039	.059	.504	-.155	.076
		6	-.147 [*]	.037	.000	-.219	-.074
2	1	2	-.267 [*]	.026	.000	-.319	-.215
		3	-.733 [*]	.046	.000	-.824	-.642
		4	-1.654 [*]	.065	.000	-1.783	-1.526
		5	-2.327 [*]	.085	.000	-2.494	-2.161
		6	-2.529 [*]	.100	.000	-2.725	-2.333
		7	-2.437 [*]	.105	.000	-2.644	-2.231
	2	1	.267 [*]	.026	.000	.215	.319
		3	-.466 [*]	.035	.000	-.535	-.397
		4	-1.387 [*]	.060	.000	-1.505	-1.270
		5	-2.060 [*]	.081	.000	-2.219	-1.901
		6	-2.262 [*]	.097	.000	-2.453	-2.070
		7	-2.170 [*]	.104	.000	-2.375	-1.966
	3	1	.733 [*]	.046	.000	.642	.824
		2	.466 [*]	.035	.000	.397	.535
		4	-.921 [*]	.051	.000	-1.022	-.821
		5	-1.594 [*]	.075	.000	-1.743	-1.446
		6	-1.796 [*]	.094	.000	-1.980	-1.611
		7	-1.704 [*]	.102	.000	-1.904	-1.504
	4	1	1.654 [*]	.065	.000	1.526	1.783
		2	1.387 [*]	.060	.000	1.270	1.505
		3	.921 [*]	.051	.000	.821	1.022
		5	-.673 [*]	.048	.000	-.768	-.578
		6	-.874 [*]	.071	.000	-1.014	-.735
		7	-.783 [*]	.082	.000	-.944	-.622
	5	1	2.327 [*]	.085	.000	2.161	2.494
		2	2.060 [*]	.081	.000	1.901	2.219
		3	1.594 [*]	.075	.000	1.446	1.743
		4	.673 [*]	.048	.000	.578	.768
		6	-.202 [*]	.045	.000	-.290	-.114
		7	-.110	.062	.075	-.231	.011
	6	1	2.529 [*]	.100	.000	2.333	2.725
		2	2.262 [*]	.097	.000	2.070	2.453
		3	1.796 [*]	.094	.000	1.611	1.980
		4	.874 [*]	.071	.000	.735	1.014
		5	.202 [*]	.045	.000	.114	.290
		7	.092 [*]	.034	.008	.024	.159
	7	1	2.437 [*]	.105	.000	2.231	2.644
		2	2.170 [*]	.104	.000	1.966	2.375
		3	1.704 [*]	.102	.000	1.504	1.904
		4	.783 [*]	.082	.000	.622	.944
		5	.110	.062	.075	-.011	.231
		6	-.092 [*]	.034	.008	-.159	-.024

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

U10.24

Cohen's d for Omnibus Main effects

Whether the simple main effect analyses support the interpretation of the main effects or not, one should, nonetheless, estimate and report the corresponding standardized effect size estimates. When there are three or more means associated with an analysis, you should consider reporting effect size estimates such as eta squared or partial eta-squared. However, when there are only two means associated with a hypothesis, in my opinion, Cohen's d (or Hedge's g) is the best way to represent the effect size. Thus, in this example, because the main effect of relationship type was based on the comparison of only two means, it may be considered especially useful to also calculate Cohen's d (or Hedge's g).

To do so with SPSS output is a little more complicated, because SPSS does not report the standard deviations associated with the main effect means. Instead, it reports only the standard errors. Fortunately, converting standard errors into standard deviations is simple with this formula: $SD = SE * \sqrt{N}$. Thus, with respect to the relationship type '1' (sexual attraction) mean, the standard deviation corresponds to a $SD = .033 * \sqrt{382} = .033 * 19.545 = .645$. Additionally, with respect to the relationship type '2' (partner desirability) mean, the standard deviation corresponds to a $SD = .035 * \sqrt{382} = .035 * 19.545 = .684$. Finally, the Cohen's d associated with the main effect of relationship type corresponds to $d = (3.696 - 3.594) / ((.645 + .684) / 2) = .102 / .664 = .154$. Again, although the relationship type omnibus effect was significant statistically ($F = 23.418, p < .001$), one would not interpret the main effect, as the simple main effect results did not support the interpretation of the main effect. Finally, I'll note that performing the standard deviation estimation calculations to three decimal places will yield respectably accurate results in virtually all cases, in practice. When the standard errors are relatively small, however, (say, less than 1.0, as per this analysis), it may be best to perform the calculations, based on values to five decimal places.

Multiple Comparisons

As was discussed in the chapter devoted to the oneway within-subjects ANOVA, there are few attractive multiple comparisons procedures available to follow-up a statistically significant main effect and which help protect from the familywise error rate from increasing unacceptably. There is simulation research to suggest that well-known multiple comparison procedures, such as Tukey-Kramer's HSD, will fail to keep familywise alpha near .05, unless the assumption of sphericity is very nearly satisfied entirely. For example, even with epsilon as large as .83 (i.e., close to 1.0; lower epsilon implies greater sphericity violation), it will impact the familywise error rate unacceptably, when conventional multiple comparison procedures are used (Boik, 1981; Maxwell, 1980). Consequently, researchers are recommended to use the Bonferroni adjustment.

$k \times k \times k$ Factorial Repeated Measures ANOVA

Like any other factorial ANOVA, it is possible to have three or more independent variables. Such designs give rise to the possibility of three-way, four-way, etc. interactions, dependent upon the number of independent included in the design. In my experience, it is usually possible to interpret a three-way interaction intelligibly with some effort. However, four-way (and greater) interactions are often beyond the researcher to grasp wholly and or communicate, effectively. I also worry about the replicability of four-way (and greater) interactions. Correspondingly, my impression is that a statistically significant four-way (and greater) interaction may be the result of a Type I error. Thus, although I am open to the possibility of four-way (and greater) interactions, I treat them with suspicion. At some point, especially if I come across an interesting one (I'm open to suggestions), I will include in this textbook an example of a study with a statistically significant three-way factorial repeated measures ANOVA interaction.

Summary

The factorial repeated measures ANOVA is a powerful design, as all of the effects (main effects and interactions) benefit from the reduction in standard error, as a consequence of the (usually) positive correlations across the conditions. The Foundations section of this chapter dealt with some of the basic procedures in the application of a factorial repeated measures ANOVA, most of which were already essentially covered in previous factorial design ANOVA chapters. A key exception was the introduction of the notion of a pooled standard error of the difference between means term, which allowed for the more accurate calculation of simple main effect t -tests. In operational terms, the adjustment involved the use of the relevant MS_E from the ANOVA table, in order to reflect the reduction in error due to the shared variance across conditions.

In the Advanced Topics section of this chapter, a number of relatively more complicated procedures are introduced. For example, the calculation of appropriate 95% confidence intervals for fully within-subjects designs. Furthermore, I show how to calculate an interaction contrast.

Advanced Topics

Adjusted Confidence Intervals: Interaction

In the chapter devoted to the test of the difference between two means, I demonstrated that typical statistical programs calculate inaccurate confidence intervals for within-subjects designs. The reason they are inaccurate is that the programs fail to take into consideration that the standard errors need to be adjusted to reflect the shared variance across the levels of the within-subjects factor (Franz & Loftus, 2012; Loftus & Masson, 1994). In the paired samples *t*-test case, there are only two levels associated with the within-subjects factor. Consequently, the method I demonstrated to obtain the adjusted standard deviations involved subtracting the covariance between the time 1 and time 2 conditions from each of the time 1 and time 2 variances. With the variances reduced in magnitude (assuming the covariance was positive, which it will almost always be), the variances were then square rooted to obtain the adjusted standard deviations. For data associated with a design more complicated than a simple paired samples *t*-test, the method of subtracting the covariance from the time 1 and time 2 variances no longer applies.⁶

Fortunately, an attractive alternative is available, based on the mean square error associated with an ANOVA summary table. The formula is simply the square root of the mean square error (MS_E). In the ANOVA context, the following formula provides the pooled adjusted standard deviation:

$$\text{pooled adjusted SD} = \sqrt{MS_E} \quad (2)$$

Recall that every factorial ANOVA includes several mean square error estimates, one for each of the *F*-tests. Consequently, with respect to the creation of accurate confidence intervals, the mean square error that should be used needs to be consistent with the level of analysis. First, I will deal with calculating adjusted standard deviations and correct confidence intervals for a 2x2 factorial repeated measures ANOVA interaction. For thoroughness, I then deal with calculating adjusted standard deviations and correct confidence intervals for the main effects.

In the 2x2 factorial repeated measures ANOVA based on student comfort levels for a professor's office, a statistically significant interaction was observed, $F(1, 250) = 16.83$, $p < .001$. The corresponding means and 95% confidence intervals are reported in the SPSS table entitled '3. Plants*cleanliness: Unadjusted'. Note that I added the 'Unadjusted' to the title myself to help distinguish the results associated with the adjusted confidence intervals I present further below.

⁶ There are also cases where the covariance is so large that it is larger than the smaller of the two variances. In such cases, the method I describe, here, could be applied to the two level (paired *t*-test) case.

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3. plants * cleanliness: Unadjusted

Measure: MEASURE_1

plants	cleanliness	Mean	Std. Error	95% Confidence Interval	
				Lower Bound	Upper Bound
1	1	6.908	.094	6.722	7.094
	2	4.418	.112	4.197	4.640
2	1	5.932	.104	5.728	6.137
	2	3.960	.110	3.744	4.177

Furthermore, as can be seen in the previously presented SPSS table entitled 'Tests of Within-Subjects Effects', the MS_E was equal to .973. Consequently, the adjusted pooled standard deviation was equal to $\sqrt{.973} = .986$. As a point of comparison, the average standard deviation across the four conditions was 1.666 ($1.495 + 1.781 + 1.647 + 1.741$). Thus, the shared variance across the four conditions reduced the variance/standard deviations considerably, within the context of the within-subjects ANOVA. Consequently, we would expect the 95% confidence intervals to be narrower, in comparison to the non-adjusted confidence intervals.

Next, I used the same technique described in the test of the difference between two means chapter: I standardized the variable scores consistent with their original means and the adjusted pooled standard deviation of .986. I then used SPSS to calculate the adjusted 95% confidence intervals via the GLM repeated measures ANOVA utility. If you have a version older than version 25, then you'll probably have to create the necessary chart with a bit of extra work (**Watch Video 10.10:** Within-Subjects Factorial ANOVA Chart with Adjusted CIs in SPSS – Less Easy). However, if you have SPSS version 25 or greater, then it will be easier to create the necessary chart (**Watch Video 10.11:** [Within-Subjects Factorial ANOVA Chart with Adjusted CIs in SPSS - Easy](#)).

As can be seen in the SPSS table entitled '3. plants*cleanliness: Adjusted', the range in the lower and upper bound 95% confidence intervals was narrower, in comparison to the unadjusted data. For example, the plants in the tidy office condition was associated with 95% confidence intervals of 6.722 and 7.094. By contrast, the adjusted data yielded adjusted confidence intervals of 6.785 and 7.031. Admittedly, the difference is rather slight, in this example, because the standard error was small even in the unadjusted data ($\approx .10$). Thus, although the adjusted standard error (.062) was about half the size of the unadjusted data, the impact on the confidence intervals was not substantial, in this case. In practice, you can expect much larger differences between the unadjusted and adjusted 95% confidence intervals.

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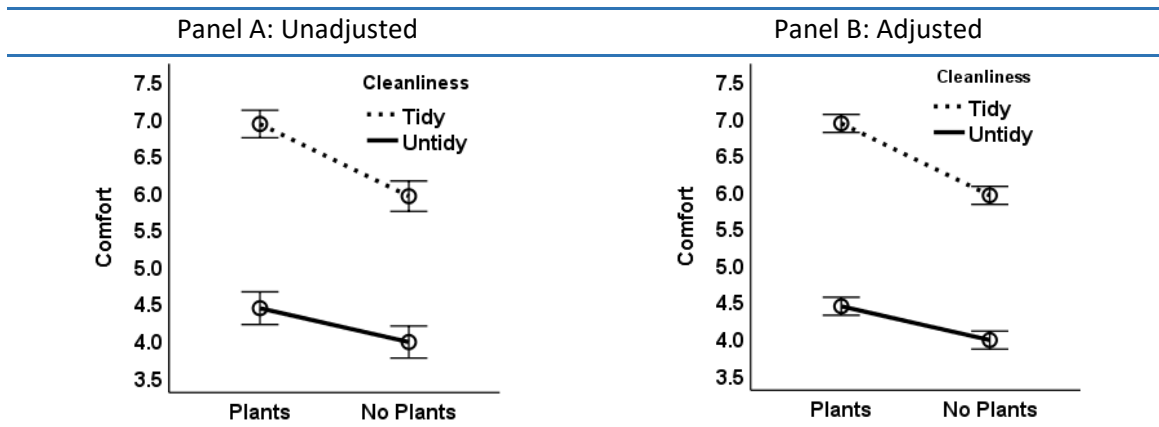
3. plants * cleanliness: Adjusted

Measure: MEASURE_1

plants	cleanliness	Mean	Std. Error	95% Confidence Interval	
				Lower Bound	Upper Bound
1	1	6.908	.062	6.785	7.031
	2	4.418	.062	4.295	4.541
2	1	5.932	.062	5.809	6.055
	2	3.960	.062	3.837	4.083

Unfortunately, for versions of SPSS prior to 25, it appears to be impossible to add confidence intervals in an efficient manner to the most attractive plot of means for a factorial repeated measures ANOVA in SPSS (see Figure C10.1). Consequently, I created another plot of the means with the 95% confidence intervals reported by SPSS, with the method described in the Foundations section of this chapter (i.e., a 'High-Low' chart in SPSS). Panel A in Figure 10.3 includes the means and the corresponding unadjusted 95% confidence intervals. Panel B includes the means and the corresponding adjusted 95% confidence intervals. Although not a very substantial difference, in this case, it is nonetheless noticeable that the adjusted confidence intervals around the means displayed in Panel B are narrower. Such an effect implies a greater level of statistical significance. Importantly, the results displayed in Panel B are more consistent with the statistically significant interaction reported in the within-subjects factor ANOVA summary table.

Figure C10.4. Plot of Means Associated with the Interaction and 95% Confidence Intervals



If you wish to report the means and confidence intervals associated with a particular main effect associated with a factorial repeated measures ANOVA, then it would be more appropriate to use the mean square error associated with that main effect, in order to adjust the corresponding standard deviations and confidence intervals. For example, in the plants and cleanliness 2x2 mixed-design ANOVA, the plants main effect was associated with a MS_E of

.973, as reported in the ANOVA summary table. That is the MS_E value you would use to conduct the necessary calculations described above.

Main Effect Means and Standard Deviations

Although it is easy to obtain the omnibus main effect means associated with a factorial repeated measures ANOVA in SPSS, it is not obvious how to obtain the standard deviations. This is disappointing, as virtually any report or manuscript should include the descriptive statistics associated with the main effects. Fortunately, with a bit of work, the standard deviations can be derived from the standard errors associated with each mean. Basically, the standard errors simply need to be back transformed into standard deviations with the following formula: $SD = SE * \sqrt{N}$. For example, as can be seen in the SPSS table entitled '1. Plants', the 'plants' versus 'no plants' means corresponded to 5.663 and 4.946. Furthermore, the standard errors corresponded to .094 and .099. Based on a sample size of $N = 251$ ($\sqrt{N} = 15.843$), the standard deviations corresponded to 1.489 and 1.568 (i.e., $SD \text{ Plants} = .094 * 15.843 = 1.489$; and $SD \text{ No Plants} = .099 * 15.843 = 1.568$).

1. plants

Measure: MEASURE_1

plants	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	5.663	.094	5.477	5.849
2	4.946	.099	4.751	5.141

The SPSS table above includes the 95% confidence intervals for each of the two means. However, these confidence intervals would not be useful for the purposes of displaying the results in a chart to compliment a repeated measures ANOVA. Instead, the unadjusted standard deviations of 1.489 and 1.568 would need to be adjusted downwardly to reflect the reduction in the standard error of the difference between the means due to the positively shared variance across the conditions. Based on the pooled adjusted standard deviation formula ($\sqrt{MS_E}$), the adjusted standard deviations associated with the two means corresponded to $\sqrt{.973} = .986$ (I obtained the MS_E value of .973 from the previously reported SPSS table entitled 'Tests of Within-Subjects Effects'). As was the case with the adjusted confidence intervals presented in Figure C10.2, the raw score data can be transformed to reflect the original means and the adjusted standard deviations. Then, SPSS can be used to calculate the 95% CIs for the adjusted data, as well as a plot of the means with the adjusted CIs as error bars.

Interaction Contrast Analyses

In the Foundations section of the chapter, the 2x2 factorial repeated measures ANOVA interaction was argued to be consistent with a statistically significant difference in the

corresponding pooled paired samples t -test values. With such knowledge, potentially interesting a sophisticated interaction contrast analyses can be performed. Additionally, the application of interaction contrast analyses can be a more powerful approach to testing hypotheses, in comparison to the conventional approach applied by typical statistical programs. First, I will demonstrate the nature of a basic interaction contrast. Then, I will demonstrate how to perform more sophisticated interaction contrasts, such as non-pairwise interaction contrasts.

The method described here to conduct an interaction contrast involves contrasting the corresponding (pooled) paired samples t -test t -values. As can be seen in the formula below, each of the t -values is accorded a weighting. The weightings need to be specified in such a way that they reflect the hypothesis. Also, like any contrast, the sum of the weightings must equal zero. In the current example, the weightings of +1 and -1 sum to zero. Furthermore, the weightings represent the hypothesis that the first pooled paired-samples t -test t -value will be larger than the second paired samples t -test t -value.

$$\text{interaction } \psi = \sqrt{\frac{((+1)(t_1) + (-1)(t_2))^2}{2}} \quad (3)$$

In the behavioral sciences, the symbol ψ typically represents a contrast. In this case, the sampling distribution associated with the contrast, under the expectation that the null hypothesis is true, follows the t -distribution with $N - 1$ degrees of freedom. In the plants and office cleanliness study described in the Foundations section of this chapter, there was a statistically significant interaction, $F(1, 250) = 17.31, p < .001$. Furthermore, the corresponding pooled paired samples t -test t -values were calculated to be 11.091 and 5.205. Thus, based on the formula above, the interaction contrast analysis yielded a contrast t -value of 4.162.

$$\text{interaction } \psi = \sqrt{\frac{((+1)(11.091) + (-1)(5.205))^2}{2}}$$

$$\text{interaction } \psi = \sqrt{\frac{(11.091 - 5.205)^2}{2}}$$

$$\text{interaction } \psi = \sqrt{\frac{34.645}{2}}$$

$$\text{interaction } \psi = 4.162$$

The interaction contrast t -value of 4.162 can be tested for statistical significance with 250 degrees of freedom (i.e., $N - 1$). When I tested the t -value with the Excel function below, I obtained a p -value of .0000436.

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=TDIST(4.162, 250, 2)	
D	E
0.0000434	

The p -value of .0000434 corresponds nearly identically to the interaction p -value obtained from the 2x2 factorial repeated measures ANOVA, $F(1, 250) = 17.306$, $p = .0000438$. I also note that the interaction contrast t -value corresponds nearly identically to the ANOVA interaction F -value, when it is known that $t^2 = F$. Thus, $4.162^2 = 17.32$. I ascribe the minor difference to rounding (17.322 versus 17.306). Thus, the interaction contrast analysis applied, here, is essentially the same as the conventional interaction analysis applied in SPSS.

	Lower-bound	195.598	250.000	.182		
plants * cleanliness	Sphericity Assumed	16.833	1	16.833	17.306	.000
	Greenhouse-Geisser	16.833	1.000	16.833	17.306	.000
	Huynh-Feldt	16.833	1.000	16.833	17.306	.0000438
	Lower-bound	16.833	1.000	16.833	17.306	.000
Error (plants*cleanliness)	Sphericity Assumed	243.167	250	.973		
	Greenhouse-Geisser	243.167	250.000	.973		
	Huynh-Feldt	243.167	250.000	.973		
	Lower-bound	243.167	250.000	.973		

The take-home message, here, is that the interaction effect tested and reported in the factorial repeated measures ANOVA summary table is a test of the difference between two (or more) t -values. In actuality, the same is true for a between-subjects factorial ANOVA. However, for the purposes of education, I tend to teach interactions in the between-subjects factorial ANOVA as essentially a test of the difference between two Cohen's d values, because it may be simpler to comprehend, in comparison to interaction contrast analyses. From a practical perspective, the advantage of having knowledge of an interaction contrast is that it allows for more flexibility in the specification of interaction hypotheses, as I describe in the next portion of the Advanced Topics section of the chapter. Additionally, an interaction contrast analysis could conceivably be applied to test the difference between two t -values obtained from two different dependent variables (i.e., dependent variables measured on different scales). I provide such an example in this chapter's practice questions.

Non-pairwise interaction contrasts (forthcoming)

Simple Main Effect Analysis: Pooled Error Term in SPSS

In the Foundations section of this chapter devoted to the plants and cleanliness 2x2 factorial repeated measures ANOVA, the simple main effect paired t -tests were performed in

C10.32

SPSS based on four separate standard error terms, i.e., the quick and dirty approach. Statistically, it is more accurate to conduct the analyses with a pooled error term, because the omnibus effects reported in the ANOVA summary table are based on a pooled error term, that is, the mean square error. Typically, pooled terms are more accurate than separately estimated terms, because pooled terms are based on more observations.

With respect to the plants omnibus main effect F -test, the mean square error was estimated at .916 (see the 2x2 mixed design ANOVA summary table reported in the Foundations section of this chapter). In the Foundations section of this chapter, I demonstrated how to estimate the pooled standard error of the difference between the means via MS_E . However, none of the simple main effect t -values derived from the pooled error term were tested for statistical significance, as testing such t -values were irrelevant to understanding the nature of an interaction.

However, if you were potentially interested in interpreting a main effect in the presence of a significant interaction, you would need to test the simple main effects for statistical significance. I demonstrate, here, how to do that with the pooled error term approach.

As I described in the Foundations section of this chapter, in order to transform the mean square error into a value that can be used as a pooled standard error of the difference between two related means, the following formula can be used:

$$\text{pooled } SE_D = \sqrt{\frac{MS_E}{(N/2)}} \quad (4)$$

Thus, with respect to the 'plants' omnibus main effect, the pooled SE_D corresponded to .085:

$$\text{pooled } SE_D = \sqrt{\frac{.916}{(251/2)}} = .085$$

With the value of .085, the two simple main effect paired t -tests were performed. In this context, the pooled error paired samples t -test corresponded to the difference between the means divided by the pooled standard error.

$$\text{pooled } t = \frac{M_1 - M_2}{\text{pooled } SE_D} \quad (5)$$

In the work developed below, you will see the two pooled simple main effect paired samples t -tests associated with the plants main effect ($t_1 = 11.482$ and $t_2 = 5.388$). Note that the isolated paired t -tests correspond to the two levels of cleanliness independent variable: tidy and untidy.

$$\begin{array}{c} \text{Tidy} \\ \text{pooled } t_1 = \frac{6.908 - 5.932}{.085} = 11.482 \end{array}$$

$$\begin{array}{c} \text{Untidy} \\ \text{pooled } t_2 = \frac{4.418 - 3.960}{.085} = 5.388 \end{array}$$

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Next, each of the pooled paired t -tests can be tested for statistical significance with $N - 1$ degrees of freedom (i.e., $251 - 1 = 250$). With the TDIST Excel function, I obtained the following results:

Tidy		Untidy	
=TDIST(11.482, 250, 2)		=TDIST(5.388, 250, 2)	
D	E	D	E
0.000000000		0.000000165	

Thus, both pooled simple main effect paired samples t -tests were statistically significant, $p < .001$. Furthermore, the t -values were in the same direction (positive). Therefore, the plant versus no plant main effect can be interpreted, in this case, even though the interaction was also significant. Such a result corroborates the separately estimated error term paired t -tests reported in the Foundations section of this chapter (i.e., quick and dirty method). However, such corroboration will not always be observed. Ultimately, the results derived from the pooled error term approach should be trusted more.

With respect to the office cleanliness omnibus main effect F -test, the mean square error was estimated at .782 (again, see the 2x2 mixed design ANOVA summary table reported in the Foundations section of this chapter). Thus, the pooled SE_D corresponded to .079:

$$\text{pooled } SE_D = \sqrt{\frac{.782}{(251/2)}} = .079$$

Next, the two pooled simple main effect paired samples t -test associated with the cleanliness main effect were performed. Note that the isolated paired t -tests corresponded to the two levels of the plants independent variable: plants and no plants.

Plants	No plants
$\text{pooled } t_1 = \frac{6.908 - 4.418}{.079} = 31.519$	$\text{pooled } t_2 = \frac{5.932 - 3.960}{.079} = 24.962$

Again, each of the pooled paired t -tests were tested for statistical significance with $N - 1$ degrees of freedom (i.e., $251 - 1 = 250$). With the TDIST Excel function, I obtained the following results:

Tidy		Untidy	
=TDIST(31.519, 250, 2)		=TDIST(24.962, 250, 2)	
D	E	D	E
0.000000000		0.000000000	

Thus, both pooled simple main effect paired samples t -tests were statistically significant, $p < .001$. Such a result corroborates the separately estimated error term paired t -tests reported in the Foundations section of this chapter. Again, such corroboration will not always be observed. When there is a difference in the results, I would put more faith in the accuracy of the pooled error term method demonstrated here. Arguably, we should always

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conduct the simple main effect analyses, based on the pooled error term, as the extra work is not substantial.

Practice Questions

Like most any statistical analyses, conducting a factorial repeated measures ANOVA involves executing a series of steps. In the following video, I describe the steps (**Watch: Factorial Repeated Measures ANOVA - Steps**). I encourage you to become familiar with the steps, prior to attempting the practice questions.

Practice Question 1: Is the Magnitude of Practice Effects Equal Across Tasks?

Practice effects on tasks are commonly observed in the behavioral sciences, when the same task, or even the same type of task, is administered to the participants on two or more occasions. A researcher wishes to determine whether practice effects are equally strong for simple tasks and complex tasks. Thus, the researcher is interested in an interaction hypothesis, i.e., the magnitude of the practice effect depends on the type of task. In this case, the challenge with testing such a hypothesis by applying something like repeated measures factorial ANOVA is that the simple and complex tasks are measured on different scales. That is, the simple task was measured in seconds to complete, whereas the complex task was measured in number of questions answered correctly. How can the researcher's interaction hypothesis be tested statistically in such a case?

One approach would be to apply formula (3) from this chapter, which is a test of the difference between two t -values (t -values are standardized; consequently, the scale of the dependent variable doesn't matter). With the data file I created for this example ([Data File: practice_effects](#)), test the following null hypotheses:

- 1: the time 1 and time 2 simple task means are equal. What is the t -value and p -value?
- 2: the time 1 and time 2 complex task means are equal. What is the t -value and p -value?
- 3: the simple task and the complex task are equally affected by practice effects. What is the t -value and p -value?

Hypothesis 1 and 2 can be tested with a paired samples t -test. Hypothesis 3 can be tested with formula (3) in this chapter.

Practice Question 2: Intelligence and Attractiveness

Apparently, intelligence is one of the most highly rated characteristics in a romantic partner. But do people really value very high levels of intelligence, when it comes to feeling attracted to another person romantically? If it were really true that intelligence was a highly rated characteristic, professors would be some of the hottest people on the planet. However, a lot of anecdotal evidence suggests that is not the case. To examine the issue, a researcher obtained data from 180 adult participants who responded to two series of seven questions on

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a 6-point Likert scale (from 1 = extremely uninterested to 6 = extremely interested). The first series of questions pertained to the degree of sexual interest (sexual relationship only) associated with a hypothetical physically attractive person who scored at the 1st, 10th, 25th, 50th, 75th, 90th, and 99th percentile in IQ. The second series of questions pertained to the degree of romantic interest (genuine relationship/possibility for marriage) associated with a hypothetical physically attractive person who scored at the 1st, 10th, 25th, 50th, 75th, 90th, and 99th percentile in IQ. Answer the following questions with respect to statistical significance ([Data File: IQ_desirability](#)):

- 1: Was there a main effect for type of relationship (sexually only versus genuine relationship)?
- 2: Was there a main effect for level of intelligence?
- 3: Was there an interaction between type of relationship and level of intelligence.
- 4: If a statistically significant interaction is observed, conduct a series of 2x2 factorial repeated measures ANOVA to isolate where the statistically significant interaction has occurred across the levels of the IQ percentiles. What were the interaction F -values associated with the six contiguous (side by side) 2x2 mixed-design ANOVAs?
- 5: Which ones were statistically significant?

The following three practice questions are very similar, although one important characteristic associated with the data is different across the three data files. By completing the three practice questions, it should become clear to you the power of a within-subjects design, as well as the true nature of an interaction. Finally, it should become clearer why it is important to conduct simple main effect analyses with a pooled error term (whether to determine if a statistically significant omnibus main effect can be interpreted, or to help explain the nature of an interaction).

Practice Question 3: No Correlations

Conduct a 2x2 factorial repeated measures ANOVA on the data file entitled '2x2_none' ([Data File: 2x2_none](#)). Answer the following questions (to 3 decimal places, minimally):

- 1: What was the omnibus interaction Mean Square Error (MS_E)?
- 2: What was the pooled Adjusted SD?
- 3: What was the pooled standard error of the difference between the means (pooled SE_D)
- 4: What were the simple main effect pooled paired t -tests (compare condition1_a with condition1_b and condition2_a with condition_2b)?
- 5: Based on the results associated with Q4, calculate the interaction contrast t -value.
- 6: Test the t -value associated with Q5 for statistical significance using the =TDIST function in Excel.

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- 7: What were the simple main effect separate (non-pooled) paired t -tests (compare condition1_a with condition1_b and condition2_a with condition_2b)?
- 8: What were the correlations between the four conditions?

Practice Question 4: Moderate Correlations

Conduct a 2x2 factorial repeated measures ANOVA on the data file entitled '2x2_moderate' ([Data File: 2x2_moderate](#)). Answer the following questions (to 3 decimal places, minimally):

- 1: What was the omnibus interaction Mean Square Error (MS_E)?
- 2: What was the pooled Adjusted SD?
- 3: What was the pooled standard error of the difference between the means (pooled SE_D)
- 4: What were the simple main effect pooled paired t -tests (compare condition1_a with condition1_b and condition2_a with condition_2b)?
- 5: Based on the results associated with Q4, calculate the interaction contrast t -value.
- 6: Test the t -value associated with Q5 for statistical significance using the =TDIST function in Excel.
- 7: What were the simple main effect separate (non-pooled) paired t -tests (compare condition1_a with condition1_b and condition2_a with condition_2b)?
- 8: What were the correlations between the four conditions?

Practice Question 5: Unequal Correlations/Standard Errors

Conduct a 2x2 factorial repeated measures ANOVA on the data file entitled '2x2_moderate_unequal' ([Data File: 2x2_moderate_unequal](#)). Answer the following questions (to 3 decimal places, minimally):

- 1: What was the omnibus interaction Mean Square Error (MS_E)?
- 2: What was the pooled Adjusted SD?
- 3: What was the pooled standard error of the difference between the means (pooled SE_D)
- 4: What were the simple main effect pooled paired t -tests (compare condition1_a with condition1_b and condition2_a with condition_2b)?
- 5: Based on the results associated with Q4, calculate the interaction contrast t -value.
- 6: Test the t -value associated with Q5 for statistical significance using the =TDIST function in Excel.
- 7: What were the simple main effect separate (non-pooled) paired t -tests (compare condition1_a with condition1_b and condition2_a with condition_2b)?

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8: What were the correlations between the four conditions?

Practice Questions: Answers

(Coming Soon)

References

- Campbell, D. E. (1979). Interior office design and visitor response. *Journal of Applied Psychology*, 64(6), 648-653.
- Loftus, G. R., & Masson, M. E. (1994). Using confidence intervals in within-subject designs. *Psychonomic Bulletin & Review*, 1(4), 476-490.
- Franz, V. H., & Loftus, G. R. (2012). Standard errors and confidence intervals in within-subjects designs: Generalizing Loftus and Masson (1994) and avoiding the biases of alternative accounts. *Psychonomic Bulletin & Review*, 19(3), 395-404.